

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(7,637,700)	(7,205,377)	6,000,000	5,660,377
2	6.258%	6.516%	(7,637,700)	(6,764,578)	6,516,291	5,771,365
3	6.522%	7.053%	(7,637,700)	(6,318,923)	7,052,703	5,834,936
4	6.794%	7.614%	(7,637,700)	(5,871,851)	7,613,820	5,853,492
5	7.075%	8.205%	0	0	0	0
				(26,160,728)		23,120,170
	Receiver strike		7.638%		Receiver	3,040,559
	Fixed rate		7.638%			
	Notional		100,000,000	NPV	0	

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	6.000%	6.000%	0x1		d_1	-0.1156
2	6.250%	6.258%	1x2	0.8857	d_2	-0.4216
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688		
5	7.000%	7.075%	4x5			
F	7.035%		$A =$	2.4818		
σ	30.6%				$XN(-d_2) - FN(-d_1)$	0.012251
T	1		Notional	100,000,000	Receiver swaption	3,040,559
X	7.638%				(in bps)	304.1

Figure S.19: Calculating the 4nc1 Cancelable Swap Rate According to the Lognormal Model

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(7,592,534)	(7,162,768)	6,000,000	5,660,377
2	6.258%	6.516%	(7,592,534)	(6,724,575)	6,516,291	5,771,365
3	6.522%	7.053%	(7,592,534)	(6,281,556)	7,052,703	5,834,936
4	6.794%	7.614%	(7,592,534)	(5,837,128)	7,613,820	5,853,492
5	7.075%	8.205%	0	0	0	0
				(26,006,027)		23,120,170
	Receiver strike		7.593%		Receiver	2,885,858
	Fixed rate		7.593%			
	Notional		100,000,000	NPV	(0)	

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	6.000%	6.000%	0x1		d	-0.2600
2	6.250%	6.258%	1x2	0.8857		
3	6.500%	6.522%	2x3	0.8273	$A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2}-dN(-d)\right]$	
4	6.750%	6.794%	3x4	0.7688		
5	7.000%	7.075%	4x5		=	0.02886
F	7.035%		$A =$	2.4818	Receiver Swaption	2,885,858
σ_N	0.02144				(in bps)	288.6
T	1		Notional	100,000,000		
X	7.593%					

Figure S.20: Calculating the 4nc1 Cancelable Swap Rate According to the Normal Model

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(7,637,700)	(7,205,377)	6,000,000	5,660,377
2	6.258%	6.516%	0	0	0	0
3	6.522%	7.053%	0	0	0	0
4	6.794%	7.614%	0	0	0	0
5	7.075%	8.205%	0	0	0	0
				(7,205,377)		5,660,377
	Payer strike		7.638%		Payer	1,545,000
	Fixed rate		7.638%			
	Notional		100,000,000	NPV	0	

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0,t)		
1	6.000%	6.000%	0x1		d_1	-0.1156
2	6.250%	6.258%	1x2	0.8857	d_2	-0.4216
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688	$FN(d_1) - XN(d_2)$	0.006225
5	7.000%	7.075%	4x5		Payer swaption (in bps)	1,545,000 154.5
F	7.035%		$A =$	2.4818		
σ	30.6%					
T	1		Notional	100,000,000		
X	7.638%					

Figure S.21: Determining the Coupon in the Extendible Swap

Now note that from the perspective of the dealer the extendible swap is equivalent to

$$\begin{aligned}
 & \text{Pay fixed for 1 year at 7.638\%} + \text{Long 1 into 3 payer struck at 7.638\%} \\
 &= \text{Pay fixed for 1 year at 7.638\%} + \text{Long 1 into 3 receiver struck at 7.638\%} \\
 &\quad + \text{Pay fixed at 7.638\% in a 1x4} \\
 &= \text{Pay fixed for 1 year at 7.638\%} \\
 &\quad + \text{Pay fixed at 7.638\% in a 1x4} \\
 &\quad + \text{Long 1 into 3 receiver struck at 7.638\%} \\
 &= \text{Pay fixed for 4 years at 7.638\%} + \text{Long 1 into 3 receiver struck at 7.638\%} \\
 &= 4\text{nc1 with a coupon of 7.638\%}.
 \end{aligned}$$

Thus, this extendible swap is equivalent to a cancelable swap that has the same coupon and has

- A final maturity date that matches the final maturity of the extendible if it is extended and
- A call date that matches the date upon which the extendible may be extended.

Note that in the first equation above we have made use of Equation S.1. In the second equation above, we have just rearranged terms. And in the second to last equation, we made the realization that if we pay fixed in a spot starting swap for one year at a rate x and we also pay fixed in a 3-year swap, one year forward at the same rate x , that is the same thing as just paying fixed in a spot starting 4-year swap at a rate x .

13. a) As shown in Figure S.22, the fixed rate is 6.895%.

b) The 4-year swap rate is 6.40%, and thus the 4-year swap rate with LIBOR in arrears is higher.

c) The fixed rate in the LIBOR in arrears swap must be higher than 6.895% when we incorporate the convexity adjustment.

LIBOR							
Time	zc	Forward	in arrears	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	6.426%	(6,895,092)	(6,529,443)	6,425,703	6,084,946
2	6.012%	6.426%	6.640%	(6,895,092)	(6,135,212)	6,640,266	5,908,470
3	6.221%	6.640%	7.086%	(6,895,092)	(5,753,186)	7,085,633	5,912,171
4	6.437%	7.086%	7.553%	(6,895,092)	(5,372,509)	7,552,520	5,884,764
5	6.659%	7.553%		0	0	0	0
					(23,790,350)		23,790,350
			Fixed rate	6.895%	NPV	0	
			Notional	100,000,000			

Figure S.22: Computing the Fixed Rate in a LIBOR in Arrears Swap (Ignoring the Convexity Adjustment)

Chapter 7

1. a) The 5-year Prime-LIBOR basis swap is -270 mid-market. Since the dealer is willing to pay 5 bps from mid-market, this implies that the dealer is willing to pay Prime $- 275$ versus receiving LIBOR flat. Since the dealer will actually be receiving LIBOR $+ 15$ from the issuer, he will be willing to pay Prime $- 275 + 15$, or Prime $- 260$. Thus the value of C is 2.60%.

b) The dealer makes 5 bps running. We should be a bit careful since payments are made quarterly. Put $PMT = 5/4 = 1.25$ bps, $n = 20$, and $i = 7.00\%/4 = 1.75\%$ and solve for $PV = 20.9$ bps of the notional, or \$209,400.

If we wanted to be a bit more careful, we would recognize that the 5 bps profitability is earned on an Act/360 basis. Notice that 5 bps Act/360 is approximately $5 \text{ bps} \times 365/360 = 5.1 \text{ bps } 30/360$. Recalculating the present value with this adjustment gives a profitability of \$212,320.

2. a) Model a swap in which the dealer receives LIBOR $- 10$ bps from the issuer and pays LIBOR $+ 10$ bps. These flows alone are negative NPV to the dealer. The dealer will be buying the cap to compensate for this shortfall.

SOLUTIONS TO SELECTED PROBLEMS

Time	zc	Forward	L – 10	PV(L – 10)	L + 10	PV(L + 10)
1	6.000%	6.000%	5,900,000	5,566,038	(6,100,000)	(5,754,717)
2	6.258%	6.516%	6,416,291	5,682,797	(6,616,291)	(5,859,933)
3	6.522%	7.053%	6,952,703	5,752,202	(7,152,703)	(5,917,669)
4	6.794%	7.614%	7,513,820	5,776,612	(7,713,820)	(5,930,371)
5	7.075%	8.205%	8,105,211	5,758,754	(8,305,211)	(5,900,854)
				28,536,403		(29,363,545)
				NPV	(827,142)	
Notional		100,000,000				

Cap model									
							Forward		Caplet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars
1	6.000%	0x1	6.000%	n/a	n/a	n/a	0.000%	0.000%	0
2	6.258%	1x2	6.516%	15%	-2.084	-2.234	0.006%	0.006%	5,577
3	6.522%	2x3	7.053%	15%	-1.048	-1.260	0.104%	0.086%	85,930
4	6.794%	3x4	7.614%	15%	-0.517	-0.777	0.334%	0.257%	256,999
5	7.075%	4x5	8.205%	15%	-0.161	-0.461	0.674%	0.479%	478,636
Strike	9.008%		Notional		100,000,000				827,142
								in bps	82.7

Figure S.23: Valuing the LIBOR Flows and Solving for the Strike on the Cap

As shown in the top half of Figure S.23, the dealer receives LIBOR – 10 bps and pays LIBOR + 10 bps, creating an NPV of –\$827,142. In order to break even, we solve for the strike on the cap such that the cap is worth \$827,142. As shown in the bottom half of Figure S.23, the strike on such a cap is 9.008%.

Thus, the coupon cap should be set to 9.008% + 10 bps = 9.108%.

Time	zc	Period	Forward	σ	$d_2[X_1]$	$d_2[X_2]$
1	6.000%	0x1	6.000%	n/a	n/a	n/a
2	6.258%	1x2	6.516%	15%	1.0554	-1.0123
3	6.522%	2x3	7.053%	15%	1.0661	-0.3959
4	6.794%	3x4	7.614%	15%	1.1219	-0.0719
5	7.075%	4x5	8.205%	15%	1.1834	0.1496

Binary				
Time	$N(d_2[X_1]) - N(d_2[X_2])$	coupon	Float	PV Float
1	n/a	(10,840,099)	6,000,000	5,660,377
2	0.6987	(7,110,440)	6,516,291	5,771,365
3	0.5107	(4,855,399)	7,052,703	5,834,936
4	0.3977	(3,513,319)	7,613,820	5,853,492
5	0.3222	(2,630,717)	8,205,211	5,829,804
		(28,949,974)		28,949,974
		NPV	0	

X_1	5.500%
X_2	7.500%
C	11.491%
Notional	100,000,000

Figure S.24: Solving for the Fixed Binary Coupon

3. a) In this trade, the dealer is receiving LIBOR and paying a fixed rate $C\%$ whenever LIBOR is between 5.50% and 7.50%. Since the first LIBOR is set at 6%, the dealer will pay a coupon of $C\%$ with certainty in one year. Thereafter, the value of the binary coupon paid at time $t + 1$ is

$$\frac{C\% \times [N(d_2[X_1]) - N(d_2[X_2])]}{[1 + zc(t + 1)]^{t+1}}.$$

As shown in Figure S.24, we solve for $X = 11.491\%$.

Chapter 8

1. a) The carry and roll-down numbers are presented in Table S.1.

3. a) If we bootstrap the curve, we find that the 4-year swap rate, one year forward is 6.541%. The value of a 1 into 4 receivers struck at 6.541% is 170.4 bps. The value of a 1 into 4 receivers struck at 6% is 89.9 bps. In dollars, the value of the first leg is \$1,703,939 and the value of the second leg is \$1,797,281. Thus, we need to solve for the strike x such that the value of leg 3 is worth \$93,342. We iterate on our swaption model and find that the strike is 4.789%.

6. a) Receiving on \$1,167.98mm of a 2x3 swap at a rate of 5.712% is equivalent to receiving on \$1,167.98mm of a 3-year swap at a rate of 5.30% and paying on \$1,167.98mm of a 2-year swap at 5.11%. Paying on \$412.57mm of a 2x5 swap at a rate of 5.976% is equivalent to paying on \$412.57mm of a 5-year swap at 5.60% and receiving on \$412.57mm of a 2-year swap at 5.11%.

b) Figure S.25a shows the decomposition of the \$1,167.98mm 2x3 swap into a spot starting 3-year swap and a spot starting 2-year swap. Notice that the LIBOR legs on these two swaps cancel each other out exactly over the first

	bps Upfront		
	Carry	Roll-down	Total
2-year swap	25.0	23.8	48.8
3-year swap	50.0	46.2	96.2
4-year swap	75.0	67.3	142.3
5-year swap	100.0	87.1	187.1
	bps Running		
	Carry	Roll-down	Total
2-year swap	26.4	25.0	51.4
3-year swap	27.2	25.0	52.2
4-year swap	28.1	25.0	53.1
5-year swap	29.1	25.0	54.1

Table S.1: Carry and Roll-Down on Swaps in Upfront bps (*top half*) and in bps Running (*bottom half*), One-Year Horizon

SOLUTIONS TO SELECTED PROBLEMS

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	61,903,088	58,955,322	(58,399,139)	(55,618,228)
2	5.113%	5.226%	61,903,088	56,027,467	(61,035,841)	(55,242,536)
3	5.312%	5.712%	61,903,088	53,000,000	(66,717,542)	(57,122,025)
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
			167,982,789		(167,982,789)	
			Fixed rate	5.300%	NPV	0
			Notional	1,167,982,789		

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(59,683,921)	(56,841,829)	58,399,139	55,618,228
2	5.113%	5.226%	(59,683,921)	(54,018,935)	61,035,841	55,242,536
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
			(110,860,765)		110,860,765	
			Fixed rate	5.110%	NPV	(0)
			Notional	1,167,982,789		

Figure S.25a: Decomposing the \$1,167.98mm 2x3 Swap into a Spot-Starting 3-Year Swap and a Spot-Starting 2-Year Swap

two years. However, the fixed cash flows over the first two years, depicted in gray, do not cancel each other out exactly.

Figure S.25b shows the decomposition of the \$412.57mm 2x5 swap into a spot starting 5-year swap and a spot starting 2-year swap. Notice that the LIBOR legs on these two swaps cancel each other out exactly over the first two years. However, the fixed cash flows over the first two years, depicted in gray, do not cancel each other out exactly.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(23,103,988)	(22,003,799)	20,628,561	19,646,249
2	5.113%	5.226%	(23,103,988)	(20,911,040)	21,559,934	19,513,542
3	5.312%	5.712%	(23,103,988)	(19,781,104)	23,566,903	20,177,441
4	5.494%	6.041%	(23,103,988)	(18,654,139)	24,924,931	20,124,366
5	5.635%	6.202%	(23,103,988)	(17,564,757)	25,588,026	19,453,242
				(98,914,839)		98,914,839
		Fixed rate	5.600%	NPV	0	
		Notional	412,571,222			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	21,082,389	20,078,466	(20,628,561)	(19,646,249)
2	5.113%	5.226%	21,082,389	19,081,324	(21,559,934)	(19,513,542)
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				39,159,790		(39,159,790)
Fixed rate			5.110%	NPV	0	
Notional			412,571,222			

Figure S.25b: Decomposing the \$412.57mm 2x5 Swap into a Spot-Starting 5-Year Swap and a Spot-Starting 2-Year Swap

The net present value of the carry over the first two years is the sum of the eight numbers that are shaded in gray in these two figures, which is \$366,976.

7. a) A year has gone by, and as it stands right now, we are long a 1 into 3 payer and short a 1 into 1 payer. Given the swap curve prevailing at this time, it can be shown that the 1x4 swap rate is 5.411%, and the 1x2 swap rate is 5.205%, giving a differential of 20.6 bps. We had put on a bullish steepener at 17.7 bps and the curve has steepened, which is favorable.